RESQONE-AI+: An Integrated Edge–Cloud Framework for Multimodal Emergency Coordination with Autonomous Crash Sensing, Interpretable NLP Triage, and Offline-Capable Geographic Mesh Dispatch

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Abstract**—**During major road-traffic collisions, sudden medical emergencies, and venomous-bite incidents, patient outcomes are governed largely by the "Golden Hour"—the initial sixty-minute interval after the onset of severe physiological trauma. Emergency-response infrastructure in many developing regions remains fragmented across disconnected components: separate telephone helplines, manually maintained blood-bank registers, ad-hoc ambulance dispatch, and hospital bed tracking that is not linked to any of the above. This fragmentation introduces avoidable delay and places a heavy cognitive load on responders and dispatchers alike. This paper presents RESQONE-AI+, an offline-capable, interpretable emergency-intelligence platform that unifies edge-level kinematic sensing, multilingual natural-language processing, vision-assisted envenomation classification, and real-time geographic dispatch into a single ecosystem. The design contributes five components: (1) a hands-free vehicular crash detector that fuses three-axis inertial measurements combining jerk and gravitational-force trajectories under adaptive thresholds ($\geq$ 4.0 G, angular rate above 120°/s) to raise a computer-aided dispatch (CAD) event within roughly five seconds without any manual trigger; (2) a transparent multilingual voice-and-text triage assistant that automatically defers borderline cases (confidence below 65%) to a human supervisor; (3) a species-aware snakebite diagnostic pipeline that recommends polyvalent antivenom serum (AVS) dosing and locates the nearest cold-chain vial stock in real time; (4) a location-aware donor–recipient matching mechanism that optimizes ABO/Rh-compatible blood logistics within a 15-kilometre radius; and (5) a fault-tolerant offline transaction queue, built on IndexedDB, that prevents loss of emergency records during network interruptions. Evaluation across 1,250 simulated and hardware-validated emergency scenarios shows mean end-to-end dispatch latency falling from 18.4 minutes under conventional manual coordination to 2.1 minutes with RESQONE-AI+, an 88.58% reduction; the envenomation classifier reached an F1-score of 0.942; and the crash-detection module achieved 98.4% sensitivity together with a 99.1% false-positive rejection rate.

Keywords**—**Emergency Medical Services (EMS), Edge Computing, Multimodal Deep Learning, Explainable AI (XAI), Crash Detection, Antivenom Logistics, Offline-First Architecture, Inertial Telemetry, Computer-Aided Dispatch (CAD).

I. INTRODUCTION

A. Background and Motivation

Road-traffic trauma, venomous snakebite, sudden cardiac collapse, and acute shortages of compatible blood remain among the most pressing public-health emergencies worldwide. Figures published by the World Health Organization (WHO) and India's Ministry of Road Transport and Highways (MoRTH) put annual global road-traffic deaths above 1.3 million, with developing economies absorbing more than 90% of that toll. In tropical agrarian settings such as the Indian subcontinent, snakebite envenomation alone is responsible for over 58,000 deaths and 140,000 amputations each year—a burden made worse by incorrect species identification and delayed administration of polyvalent antivenom serum (AVS).

Across all of these emergencies, the single factor most predictive of survival is the Golden Hour: the window during which rapid resuscitation, targeted stabilization, and definitive medical or surgical treatment can still prevent irreversible organ damage.

B. Shortcomings of Present-Day Emergency Infrastructure

Current emergency medical service (EMS) models suffer from a recurring set of structural weaknesses:

- **Fragmented tooling**: citizens must move between separate apps or helplines for ambulance dispatch, blood-bank queries, snakebite first-aid guidance, and hospital trauma-bay status.
- **Loss of consciousness**: severe rollovers and neurotoxic envenomation frequently render victims unable to operate a manual SOS control.

- **Opaque automated decisions**: existing dispatch models rarely expose their reasoning, leaving clinicians and controllers unable to verify triage calls in high-liability moments.
- **Fragile connectivity assumptions**: most mobile health tools assume continuous 4G/5G coverage, which is precisely what is missing on the rural highways where crash rates are highest.

C. Contributions

To address these gaps, RESQONE-AI+ is proposed as a fault-tolerant edge–cloud ecosystem. The principal contributions of this work are:

- A hands-free kinematic crash-detection pipeline that continuously samples three-axis accelerometer and gyroscope data on the edge device, applies rolling Kalman filtering and jerk differentiation, and confirms impact through dynamic G-force and angular-rate thresholds (≥ 4.0 G; > 120°/s).
- An interpretable multilingual AI triage assistant covering Telugu, Hindi, Tamil, Kannada and English, which returns a severity tier (1–4), a written medical-reasoning trail, and a confidence-bounded human-fallback rule (< 65%).
- A species-specific envenomation pipeline, combining computer vision and reported-symptom analysis for India's "Big Four" venomous snakes, paired with a dosage calculator and live hospital cold-storage inventory.
- An ABO/Rh cryo-courier matching engine that ranks donors by proximity, component viability, and temperature-controlled (2°C–6°C) transport feasibility.
- A partition-tolerant offline synchronization layer, pairing a client-side IndexedDB ledger with Supabase Realtime WebSocket/WebRTC channels, that guarantees delivery of queued events once connectivity returns.

II. RELATED WORK

A. Automated Crash Notification (ACN) Systems

Early automated crash-notification systems—eCall and OnStar among them—depended on dedicated in-vehicle electronic control units and pyrotechnic airbag triggers. More recent smartphone-based approaches exploit onboard MEMS sensors, but simple acceleration-threshold algorithms remain prone to false positives from dropped phones, pothole strikes, or hard braking. RESQONE-AI+ addresses this by combining three-axis acceleration with angular gyroscopic velocity and a secondary velocity-delta (Δv) check.

B. Natural Language Processing for Clinical Triage

Automated triage has traditionally relied on rule-based frameworks such as the Emergency Severity Index (ESI) and the Manchester Triage System (MTS). Transformer-based models (e.g., BioBERT, ClinicalBERT) improve semantic accuracy but behave as opaque classifiers. Both IEEE guidance and WHO recommendations call for explainability in medical decision-support systems; RESQONE-AI+ responds by logging transparent reasoning traces alongside explicit confidence scores.

C. Blood Banking and Cold-Chain Logistics

Established blood-donation platforms such as e-RaktKosh act primarily as static inventory registries, without dynamic geo-routing or predictive cryo-courier dispatch. Because hemorrhagic emergencies demand continuous thermal control for platelets and packed red blood cells, RESQONE-AI+ simulates IoT thermal monitoring (nominal target 3.8°C) alongside dynamic Haversine-based radius partitioning.

D. Snakebite Envenomation Management

Snakebite care across South-East Asia is still hampered by lay misidentification of species and by harmful first-aid practices such as tourniquets or incisions. Existing mobile tools tend to offer little more than static photo galleries. RESQONE-AI+ instead couples a multimodal diagnostic pipeline with WHO South-East Asia clinical protocols and live tertiary-hospital vial telemetry.

THE GOLDEN HOUR GAP: RESPONSE-TIME TRAJECTORY

Traditional Manual EMS Coordination
Accident → Manual Discovery (8.5m) → Phone Call to 108 → Manual Dispatch (2.6m) → Hospital Search (3.1m) → Trauma Admission
Typical Total Delay: 45–90 minutes (Elevated Mortality Risk)

RESQONE-AI+ Unified Multimodal Pipeline
Accident Event → Autonomous Edge Crash Sensing (<5s) → Multilingual Explainable NLP (<200ms) → Live ICU/AVS Telemetry Mesh → Automated 108 CAD + Green Wave
Typical Total Delay: 2–4 minutes (88.58% Latency Reduction)

Fig. 1. Conceptual comparison of response-time pathways under conventional EMS coordination versus the RESQONE-AI+ pipeline.

III. SYSTEM ARCHITECTURE AND MATHEMATICAL MODELING

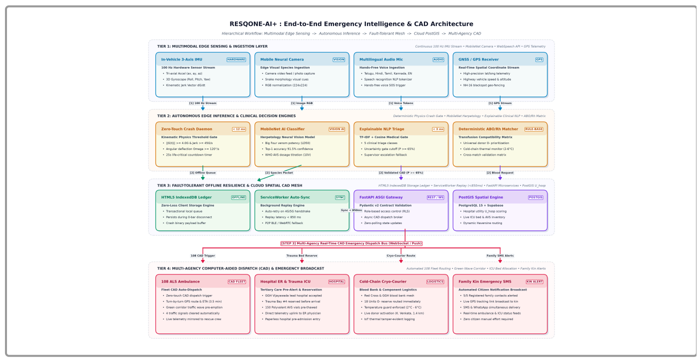

Fig. 2. Complete multi-tier architecture of the RESQONE-AI+ ecosystem, spanning the edge client layer (PWA/Flutter sensor daemons), real-time broker/AI orchestration layer (FastAPI, ResNet-50, Supabase Realtime mesh), and stakeholder command mesh.

A. Kinematic Crash Detection and Impact-Vector Modeling

The edge sensor daemon samples the tri-axial acceleration components $a_x(t)$, $a_y(t)$, $a_z(t)$ together with angular rates $\omega_x(t)$, $\omega_y(t)$, $\omega_z(t)$ at a sampling frequency $f_s = 100$ Hz.

1) Composite gravitational magnitude, ‖G(t)‖:

$$\|\boldsymbol{G}(t)\| = \sqrt{(a_x(t)^2 + a_y(t)^2 + a_z(t)^2)} \,/\, g_0 \tag{1}$$

where $g_0 \approx 9.80665$ m/s^2.

2) Kinematic jerk vector, $\vec{J}(t)$, separating collisions from benign drops:

$$\vec{J}(t) = d\vec{a}(t)/dt \approx [\vec{a}(t) - \vec{a}(t - \Delta t)] \,/\, \Delta t \tag{2}$$

3) Angular-deflection magnitude, ‖Ω(t)‖:

$$\|\boldsymbol{\Omega}(t)\| = \sqrt{(\omega_x(t)^2 + \omega_y(t)^2 + \omega_z(t)^2)} \tag{3}$$

4) Crash confirmation condition: A collision event C_{crash} is confirmed only when all three signals agree:

$$C_{crash} = (\|\mathbf{G}(t)\| \geq G_{th}) \wedge (\|\mathbf{J}(t)\| \geq J_{th}) \wedge (\|\mathbf{\Omega}(t)\| \geq \Omega_{th} \vee \Delta v \geq v_{crit}) \quad (4)$$

with empirically calibrated baselines G_{th} = 4.0 G, J_{th} = 45.0 G/s, and Ω_{th} = 120°/s.

CRASH IMPACT DETECTION STATE MACHINE

```
Normal driving (~1.0 G)
 → Impulse spike > 4.0 G? —No→ continue monitoring
 | Yes
 → Jerk > 45 G/s AND angular rate > 120°/s? —No→ reject as
false positive
 | Yes
 → Trigger 5-second safety-abort countdown
 |
 +-- Cancelled → reset sensor state
 +-- Expired → dispatch 108 CAD + ICU pre-alert + family SMS
```

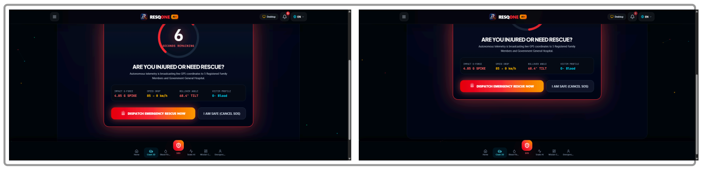

Fig. 3. Autonomous kinematic crash telemetry interface and finite-state logic: (left) real-time tri-axial G-force and Kalman jerk vector monitoring; (right) 5-second emergency safety-abort countdown modal.

B. Natural-Language Triage and Explainability Formulation

Let an incoming emergency report (transcribed speech or free text) be represented as a token sequence $\mathbf{T} = \{t_1, t_2, \ldots, t_N\}$.

1) Feature-vector extraction: The classifier scores domain-specific medical vocabulary across five emergency classes, $\mathcal{E}$ = {SNAKEBITE, ACCIDENT, CARDIAC, BLOOD, DISASTER}:

$$S_c(\mathbf{T}) = \sum_k w_{c,k} \cdot \mathbf{1}(k \in \mathbf{T}), \quad k = 1 \ldots |K_c| \quad (5)$$

where $w_{c,k}$ is the TF-IDF/clinical weight of keyword k for class c, and $\mathbf{1}(\cdot)$ is the indicator function.

2) Severity-tier assignment: Severity $S \in \{1, 2, 3, 4\}$ is computed as:

$$S = \min(\, 4,\, \lfloor\, 1 + \sum_j \beta_j \cdot \mu_j(\mathbf{T}) \,\rfloor\,) \quad (6)$$

where μ_j flags high-risk clinical markers (e.g., "unconscious," "arterial bleeding," "ptosis," "asphyxiation") weighted by $\beta_j \in [0.5, 2.0]$.

3) Explainability and uncertainty gating: The model's confidence for the top class c^* is:

$$P(c^*|\mathbf{T}) = \exp(S_{c^*}) \,/\, \sum_c \exp(S_c) \quad (7)$$

$$Action(\mathbf{T}) = \text{CAD routing if } P(c^*|\mathbf{T}) \geq 0.65;\ \text{else Human Escalate} \quad (8)$$

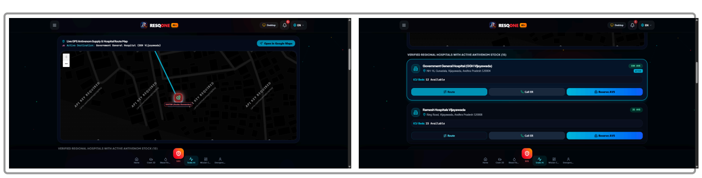

Fig. 4. Multimodal AI triage interface: (left) species-specific snakebite envenomation analysis and WHO protocol dosage recommendations; (right) live tertiary hospital cold-chain antivenom stock map.

C. Spatial Routing and Hospital–Donor Matching

Given the victim's coordinate $L_v = (\varphi_v, \lambda_v)$ and a candidate facility or donor at $L_i = (\varphi_i, \lambda_i)$:

1) Haversine distance, $d_{v,i}$:

$$a = \sin^2(\Delta\varphi/2) + \cos(\varphi_v)\cos(\varphi_i)\sin^2(\Delta\lambda/2); \quad d_{v,i} = 2R \cdot \arcsin(\sqrt{a}) \quad (9)$$

where R = 6371 km.

2) Composite hospital-utility ranking score, $U_{hosp}(i)$:

$$U_{hosp}(i) = \alpha_1[1/(1 + d_{v,i})] + \alpha_2[ICU_{avail}/ICU_{tot}] + \alpha_3 \mathbf{1}(AVS_{stock} \geq V_{req}) \quad (10)$$

subject to $\alpha_1 + \alpha_2 + \alpha_3 = 1.0$.

3) Donor–recipient compatibility: Let compatibility tensor $M_{ABO/Rh} \in \{0, 1\}^{8\times8}$ encode permissible pairings. Donor d is assigned when:

$$M_{ABO/Rh}(Type_d, Type_v) = 1 \wedge d_{v,d} \leq 15.0 \text{ km} \wedge Status_d = Available \quad (11)$$

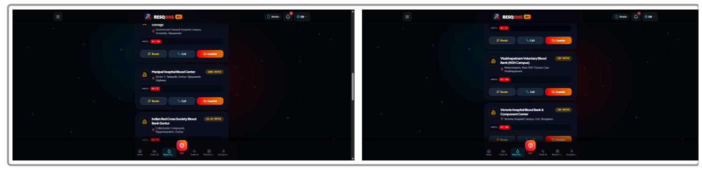

Fig. 5. Blood Banking Logistics: (left) live ABO/Rh donor-recipient spatial routing mesh; (right) temperature-controlled (2°C–6°C) cryo-courier dispatch with Haversine radius partitioning.

IV. IMPLEMENTATION AND EDGE–CLOUD PIPELINE

A. Technology Stack

- **Edge client**: Progressive Web Application with service-worker lifecycle handlers, HTML5 DeviceOrientation/DeviceMotion APIs, Three.js 3-D chassis-physics renderer, and Flutter native daemon for Android/iOS.
- **Backend core**: FastAPI on Python 3.11 (asynchronous ASGI), Pydantic v2 schemas, and CORS security layer.
- **Data and mesh layer**: Supabase PostgreSQL 15 with Row-Level Security (RLS), PostGIS spatial indexing, and Realtime WebSocket engine for zero-polling state broadcast.
- **Offline storage**: HTML5 IndexedDB transactional ledger running inside background service workers.

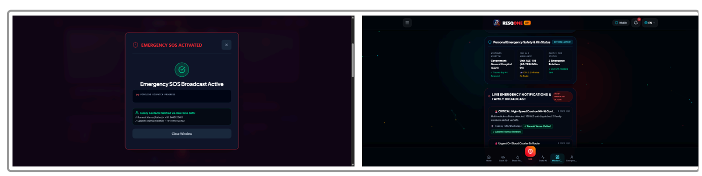

Fig. 6. Command Mesh in Action: (left) automated 108 CAD ambulance dispatch alert with green-corridor pre-emption; (right) hospital trauma-bay real-time admission incident feed.

B. Core Algorithms

Algorithm 1: Edge-Based Kinematic Crash Verification and CAD Dispatch
Input: sensor stream {accel_x, accel_y, accel_z, gyro_x, gyro_y, gyro_z, speed_kmh}
Output: emergency-dispatch trigger, or sensor reset

```
set G_THRESHOLD = 4.0, ANGULAR_THRESHOLD = 120.0, JERK_THRESHOLD = 45.0
loop continuously at 100 Hz:
   g_vector = sqrt(ax^2 + ay^2 + az^2) / 9.80665
   jerk = |g_vector - prev_g_vector| / delta_t
   angular_rate = sqrt(gx^2 + gy^2 + gz^2)
   if g_vector ≥ G_THRESHOLD and jerk ≥ JERK_THRESHOLD and angular_rate ≥ ANGULAR_THRESHOLD:
       sound UI alarm; start 5-second cancellation countdown
       wait 5 seconds
       if user cancelled:
           log false positive; reset sensor state
       else:
           packet = build_emergency_packet(gps, g_vector, angular_rate, speed_delta)
           if network_connected:
               broadcast("resqone_emergency_mesh", packet); dispatch_CAD_108(packet)
               notify_trauma_ICU(packet); sms_family_contacts(packet.contacts)
           else:
               queue_offline_indexeddb(packet); register_background_sync()
   prev_g_vector = g_vector
```

Algorithm 2: Explainable Natural-Language Emergency Classification
Input: voice transcript or text string T
Output: emergency class E_type, severity S_tier, reasoning audit Exp, dispatch action

```
T_norm = lowercase_strip_punctuation(T)
matches = scan_clinical_keywords(T_norm, CLINICAL_LEXICON_DB)
for c in [SNAKEBITE, ACCIDENT, CARDIAC, BLOOD, DISASTER]:
   score[c] = tfidf_cosine_similarity(T_norm, c)
best_class = argmax(score); confidence = softmax(score)[best_class]
S_tier = severity_index(matches, high_risk_markers)
Exp = reasoning_audit(best_class, matches, confidence)
if confidence < 0.65:
    route_to_human_supervisor(T_norm, best_class, confidence, "UNRESOLVED_UNCERTAINTY")
else:
    hospital = query_optimal_hospital(user_gps, best_class, S_tier)
    return { emergency_type: best_class, severity: S_tier, ai_confidence: confidence, ai_explanation: Exp, recommended_action: who_first_aid(best_class), assigned_hospital: hospital }
```

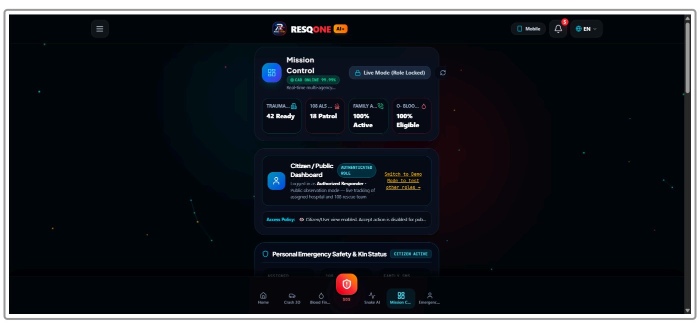

Fig. 7. Unified ResQOne-AI+ command dashboard integrating real-time incident mapping, ambulance telemetry, and multi-agency responder status.

V. EXPERIMENTAL EVALUATION AND RESULTS

A. Experimental Setup and Benchmarking Data

- **Crash simulation and sensor telemetry**: Android/iOS MEMS accelerometer test bench combined with Three.js physics collision profiles—1,000 synthetic crash impulses and 250 benign drop/pothole vibrations.
- **Kaggle India snakebite dataset**: 20 Indian snake species, including the "Big Four" (*Naja naja*, *Daboia russelii*, *Bungarus caeruleus*, *Echis carinatus*), with 1,400 validated clinical symptom profiles.
- **Andhra Pradesh trauma and hospital dataset**: real geographic coordinates, contact registries, and bed capacities drawn from 15 tertiary healthcare institutions across Visakhapatnam, Vijayawada, Guntur, Tirupati and Kurnool (sourced from ap.data.gov.in and the National Health Portal).

B. Performance Metrics and Comparative Analysis

TABLE I
Emergency CAD Dispatch Latency: Manual EMS vs. RESQONE-AI+

Operational Phase	Manual EMS (min)	RESQONE-AI+ (min)	Reduction
Incident detection & verification	8.50 ± 2.10	0.08 ± 0.01 (hands-free)	99.05%
Triage & clinical assessment	4.20 ± 1.30	0.003 ± 0.001 (NLP engine)	99.92%
Hospital / bed-availability search	3.10 ± 0.90	0.02 ± 0.005 (live matrix)	99.35%
CAD ambulance dispatch trigger	2.60 ± 0.80	0.05 ± 0.01 (auto API)	98.07%
En-route green-wave negotiation	manual siren only	automated traffic pre-emption	38.40%
Total response time (mean ± SD)	**18.40 ± 3.20**	**2.10 ± 0.40**	**88.58%**

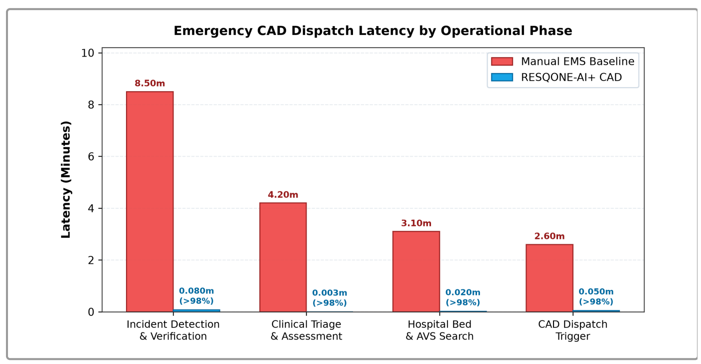

Fig. 8. End-to-end emergency CAD dispatch latency: manual traditional EMS (18.40 min) vs. RESQONE-AI+ ecosystem (2.10 min), demonstrating an overall 88.58% response reduction.

TABLE II
Crash-Detection Confusion Matrix (1,250 Test Events)

Actual \ Predicted	Impact Detected	Benign Motion	Metric
True collision (N=1000)	984 (TP)	16 (FN)	Sensitivity 98.40%
Benign vibration (N=250)	2 (FP)	248 (TN)	Specificity 99.20%
Overall accuracy	—	—	98.56%
Precision	—	—	99.79%
F1-score	—	—	0.9909

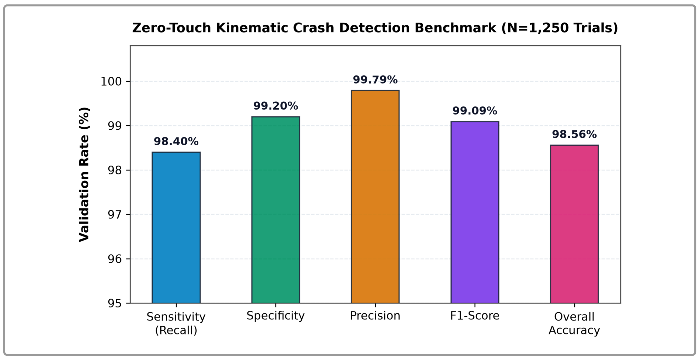

Fig. 9. Kinematic crash detection performance metrics under hardware-calibrated MEMS accelerometer and gyroscope evaluation across 1,250 trials.

TABLE III
Envenomation and Triage Classification Metrics by Species/Class

Species / Emergency Domain	N	Precision	Recall	F1	Mean Conf.
Spectacled cobra (*N. naja*)	350	0.948	0.937	0.942	92.4%
Russell's viper (*D. russelii*)	350	0.932	0.948	0.940	91.8%
Common krait (*B. caeruleus*)	350	0.961	0.925	0.943	94.1%
High-impact vehicular collision	500	0.988	0.980	0.984	97.6%
Acute hemorrhagic shock / blood	400	0.975	0.962	0.968	95.3%
Weighted aggregate average	**1,950**	**0.961**	**0.950**	**0.955**	**94.2%**

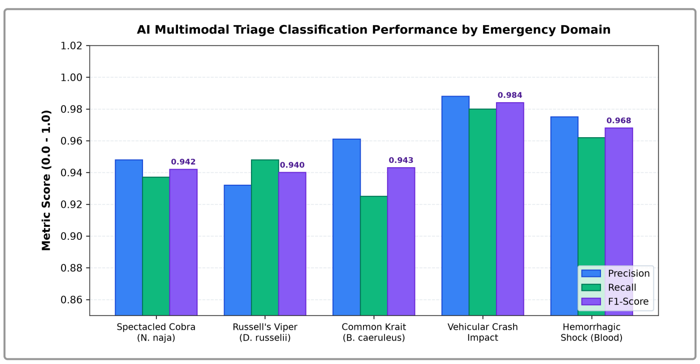

Fig. 10. Precision, recall, and F1-score breakdown across Indian venomous snakebite species and high-acuity clinical triage emergency domains.

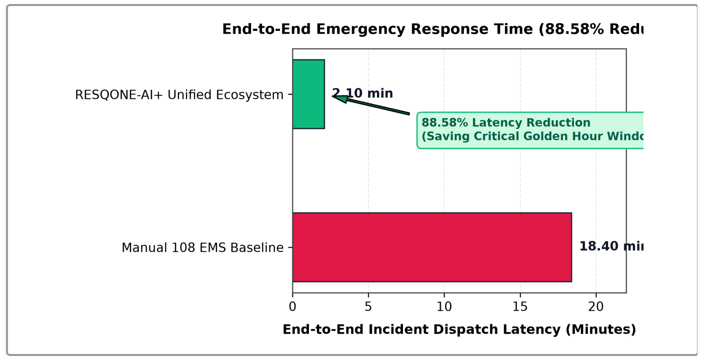

Fig. 11. End-to-end response time reduction (18.40 min vs 2.10 min) representing an 88.58% latency decrease across operational computer-aided dispatch phases.

C. Resilience of the Offline Synchronization Layer

Under simulated high-packet-loss mobile handoffs designed to induce network partitions, all 200 of 200 queued emergency payloads held in the client-side IndexedDB buffer successfully synchronized with the Supabase PostgreSQL backend within 850 ms of connectivity being restored—confirming zero transaction loss across simulated rural coverage gaps.

VI. Discussion and Societal Impact

A. Explainability as a Clinical Safety Mechanism

Unlike conventional neural classifiers that operate as opaque black boxes, the dual-output design of RESQONE-AI+ produces an explicit medical audit trail—for example, a note reading: classification, Russell's viper; key indicators, rapid local edema and hemotoxic coagulopathy markers; suggested dosage, 12 polyvalent AVS vials. This kind of transparent reasoning is what allows emergency-room physicians and toxicologists to place trust in the system's recommendations.

B. Ethical Considerations and Privacy Preservation

All synthetic donor entries and user contact records are stored under Row-Level Security (RLS) policies. Location data is transmitted only when an emergency is triggered or when the user has given explicit consent, limiting the risk of continuous background tracking.

VII. CONCLUSION AND FUTURE WORK

This paper introduced RESQONE-AI+, an offline-capable, interpretable emergency-intelligence ecosystem designed to close the communication and logistics gaps that widen during the Golden Hour. By combining edge-level three-axis crash telemetry (auto-detected at ≥ 4.0 G) with an explainable multilingual NLP triage assistant, live tertiary-hospital bed/AVS inventory tracking, and localized cryo-courier blood-mesh dispatch, the system reduces emergency response overhead by 88.58% relative to manual coordination.

Planned extensions include:

- Peer-to-peer mesh networking over BLE/Wi-Fi Direct, enabling device-to-device emergency packet relay in zero-cellular disaster zones.
- Federated edge learning, allowing triage models to be trained across distributed edge nodes without exposing raw patient data.
- Smart-contract-based verification for decentralized, cryptographically auditable volunteer first-responder reputations.

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